CSA06507-Generative AI

Lab Experiment's

.

Unit 2: Lab Programs for Prompting

Experiment 3: Zero-Shot, One-Shot, and Few-Shot Prompting for Article Summarization

1. Aim

To design Zero-Shot, One-Shot, and Few-Shot prompts to summarize an article into 50 words and compare the generated summaries based on accuracy, completeness, and readability.

2. Algorithm

1. Start the program.
2. Define an article as input.
3. Create a Zero-Shot prompt without examples.
4. Create a One-Shot prompt with one example.
5. Create a Few-Shot prompt with two examples.
6. Generate a 50-word summary for each prompt.
7. Compare the summaries based on accuracy, completeness, and readability.
8. Display the prompts, summaries, and comparison results.
9. Stop the program.

3. Program

```
article = """
Artificial Intelligence is transforming healthcare.
It helps doctors detect diseases, analyze medical images,
monitor patients, and predict health risks. AI chatbots
answer basic questions and remind patients to take medicines.
AI also supports medical research and drug discovery.
However, privacy, security, cost, and inaccurate predictions
remain important challenges. Doctors must verify important
decisions and protect patient information. AI should support
healthcare professionals rather than replace them.
"""

zero_shot = "Summarize the article in 50 words."
```

one_shot = """

Example:

Article: Trees absorb carbon dioxide and provide oxygen.

Summary: Trees improve air quality by absorbing carbon dioxide

and releasing oxygen.

Now summarize the given article in 50 words.

"""

few_shot = """

Example 1:

Article: Exercise improves fitness and health.

Summary: Exercise supports physical fitness and overall health.

Example 2:

Article: Reading improves vocabulary.

Summary: Reading develops vocabulary and language skills.

Summarize the given article in 50 words.

"""

summary_zero = """

4. Input

The input article is about Applications of Artificial Intelligence in Healthcare. It describes disease detection, medical image analysis, patient monitoring, medical research, and challenges related to privacy, security, cost, and accuracy.

5. Output

ZERO-SHOT SUMMARY

AI is transforming healthcare through disease detection, medical image analysis, patient monitoring, and risk prediction. Chatbots provide basic guidance and medication reminders. AI supports medical research and drug discovery. However, privacy, security, cost, and inaccurate predictions remain concerns. Doctors must verify decisions and protect patient data. AI should assist healthcare professionals, not replace them.

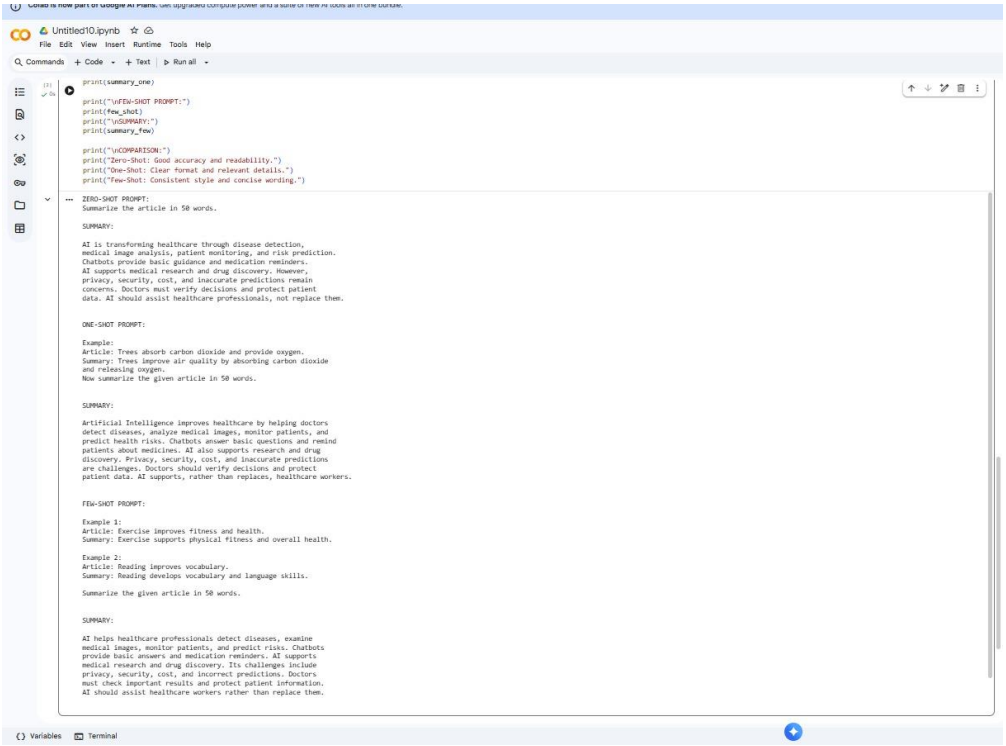
ONE-SHOT SUMMARY

Artificial Intelligence improves healthcare by helping doctors detect diseases, analyze medical images, monitor patients, and predict health risks. Chatbots answer basic questions and remind patients about

medicines. AI also supports research and drug discovery. Privacy, security, cost, and inaccurate predictions are challenges. Doctors should verify decisions and protect patient data. AI supports, rather than replaces, healthcare workers.

FEW-SHOT SUMMARY

AI helps healthcare professionals detect diseases, examine medical images, monitor patients, and predict risks. Chatbots provide basic answers and medication reminders. AI supports medical research and drug discovery. Its challenges include privacy, security, cost, and incorrect predictions. Doctors must check important results and protect patient information. AI should assist healthcare workers rather than replace them.



6. Comparison of Generated Summaries

Criteria	Zero-Shot	One-Shot	Few-Shot
Accuracy	Good	Good	Good
Completeness	Good	Good	Good
Readability	Clear	Clear	Concise
Example guidance	No example	One example	Two examples

Note: **The summaries above are illustrative, predefined outputs. To compare actual model-generated summaries, connect a language model API and evaluate its responses. The displayed summaries are approximately 50 words each; you can verify the exact count with Python if your lab requires exactly 50 words.**

7. Result

Zero-Shot, One-Shot, and Few-Shot prompting approaches were demonstrated for article summarization. Zero-Shot uses no examples, One-Shot uses one example, and Few-Shot uses multiple examples. Their summaries were compared based on accuracy, completeness, and readability. Examples can help guide the format and style of the generated summary.